Causal effects of schools on higher education outcomes - Spring update

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1 Introduction

In this report, I present the progress I have made in the last month. I will go over the intuition behind the main changes in my approach and the current methodology I am using. I will show some results and briefly discuss their interpretation.

2 From school-specific effects to a scalar school-value treatment.

My original intent of this paper was to measure idiosyncratic effects of each school on higher education outcomes. The main challenge I was facing as a result, was that of having many unordered treatments – one per high school and the absence of a clear control group. The school assignment system is designed to give all students a school[1]. Even if some students drop out (and we ignore selection into that), we would need individuals dropping out on each 'randomization cluster'.

As an example, suppose a mass of students with identical preferences for 3 schools, say $A \succ B \succ C$, and identical priority levels. Suppose furthermore, that all of them have guaranteed admission to school G once the spots have run out for these three schools. Because we could compare the outcome of each school relative to the guaranteed admission school G, we could estimate the effect of each school relative to G.

We could construct something like $\beta_{AG}, \beta_{BG}, \beta_{CG}$, where β_{AG} would be the effect of attending school A relative to G, and so on. Given the common comparison group, we could even go further and call these effects $\beta_A, \beta_B, \beta_C$, and even rank them (e.g. $\beta_A > \beta_B > \beta_C \implies A > B > C$) as they would be comparable across schools.

Suppose now that these students have guaranteed admission to different schools, say $G = (G^1, G^2, ..., G^{10})$. Note now that we would need to estimate 30 objects $(\beta_{AG^1}, \beta_{AG^2}, ..., \beta_{CG^{10}})$

[1] no student is left unassigned: if unlucky in the first round students are assigned a spot in their original schools, and a complementary round matches students to the closest available spot at the end

and that we would not be able to rank them as they are not comparable across schools, except for comparisons within the same guaranteed admission school. In this case, we would not be able to say anything about the relative ranking of schools A, B, and C.

One solution to this problem is to collapse some of these treatment options. For instance, collapse guaranteed admission schools into two buckets, and define $\tilde{G} = (G^L, G^H)$. An example of something similar is Kirkeboen et al. (2016), which studies the causal effect of different higher education programs on labor market outcomes. Given that they use a centralized assignment mechanism for higher education in Norway, their instrument is the score cutoff for each program. Given that people have different lists, they have different fallback options. Instead of estimating all the program effects relative to each other, they collapse the programs only into 10 fields and estimate 90 next-best effects. Note that to collapse the treatment options, they abstract away from differences in institutional quality in their main specification.

Table 5. 2SLS estimates of the payoffs to field of study (USD 1,000)

	Next best alternative (k):								
	Humanities	Soc Science	Teaching	Health	Science	Engineering	Technology	Business	Law
Completed field (j):									
Humanities		21.38*	-4.72	-22.93*	4.97	-38.51**	6.87	-42.21**	-156.33
		(10.97)	(9.85)	(12.12)	(11.86)	(14.72)	(48.29)	(10.56)	(437.28)
Social Science	18.72**		9.84	-10.82	55.46**	-55.36**	-110.38	-28.37**	-76.07
	(6.73)		(11.55)	(13.00)	(21.45)	(20.60)	(102.97)	(10.66)	(86.42)
Teaching	22.25**	31.37**		1.82	23.46**	-33.94**	-35.32	-21.08**	22.78
	(4.96)	(7.88)		(6.55)	(9.45)	(12.54)	(37.07)	(7.12)	(127.87)
Health	18.75**	30.69**	7.72**		28.87**	-27.87**	-43.38**	-17.39**	-55.19
	(6.25)	(7.56)	(2.82)		(7.64)	(10.35)	(20.84)	(3.97)	(97.68)
Science	53.71**	69.59**	38.58**	29.63**		-2.21	16.81	-4.92	148.26
	(18.37)	(22.36)	(14.20)	(11.53)		(14.60)	(18.07)	(10.51)	(276.20)
Engineering	59.81	-5.53	75.24**	0.16	52.35**		-46.00	-13.03	-57.66
	(50.59)	(58.17)	(37.50)	(16.36)	(20.98)		(43.89)	(23.70)	(166.60)
Technology	41.87**	58.69**	22.08*	32.45**	68.07**	-5.56		7.03	-53.07
	(10.84)	(10.09)	(12.44)	(10.09)	(9.63)	(11.95)		(9.49)	(147.53)
Business	48.13**	61.93**	31.02**	30.22**	58.01**	-3.42	28.54*		3.53
	(11.25)	(12.03)	(8.78)	(10.86)	(10.48)	(12.61)	(15.61)		(83.04)
Law	46.34**	55.62**	36.60**	21.49*	40.07**	-27.53	-15.55	-1.36	
	(7.16)	(8.34)	(11.56)	(11.46)	(9.68)	(18.29)	(17.96)	(8.66)	
Medicine	83.34**	79.39**	62.62**	45.57**	81.31**	21.07	40.07**	23.34**	14.82
	(9.76)	(10.65)	(9.02)	(7.01)	(9.71)	(20.67)	(11.72)	(8.79)	(83.61)
Average y^k	30.01	23.40	46.15	51.79	27.31	87.85	78.37	75.61	105.83
Observations	8,391	11,030	10,987	3,269	6,422	3,085	1,245	4,403	1,251

Note: From 2SLS estimation of equations (14)-(15), we obtain a matrix of the payoffs to field j as compared to k for those who prefer j and have k as next-best field. Each cell is a 2SLS estimate (with st. errors in parenthesis) of the payoff to a given pair of preferred field and next-best field. The rows represent completed fields and the columns represent next-best fields. The row labeled average y^k reports the weighted average of the levels of potential earnings for compliers in the given next-best field. The final row reports the number of observations for every next-best field. Stars indicate statistical significance, * 0.10, ** 0.05.

Figure 1: Effect of payoffs of different fields of study relative to their next-best alternative. Source: Kirkeboen et al. (2016).

My original concern with doing something like this in my paper was my intention to use this paper to identify whether specific schools were overperforming their peer institutions on some outcomes, as opposed to asking if some group of schools were overperforming others.

Talking to Prof. Walters was especially helpful in realizing that instead of collapsing schools on a categorical variable, I could do so on a continuous one. More importantly, the continuous variable can be constructed in a way that preserves the idiosyncratic effects of each school, while also allowing for comparability across schools.

Going back to our original example, if we have a metric V for each of the schools (e.g. $V(A) = 3, V(B) = 5, V(C) = 7, V(G^i) = i$) then we can pool over multiple comparisons. Outcome differences between A relative to G^1 (a gain of 2 in the V metric) would be comparable to outcome differences between B relative to G^3 (also a gain of 2 in the V metric), and so on. This way, we can estimate a single parameter θ that captures the effect of attending a school with a higher V metric relative to attending a school with a lower V metric.

In particular, I will construct these V metrics using the observational value-added of each school on a particular outcome. I will then answer the question whether movements in a particular metric induced by the lottery offers, are correlated with differences in outcomes. Given that the lottery offers encode exogenous variation we can interpret the results causally.

3 Methodology

3.1 Computing Observational Value-Added for each school

Following the value-added literature (Chetty et al. 2014; Angrist et al. 2017), I run a regression of the following form to compute the observational value-added for each school for each of these outcomes:

$$Y_i = \mu + \beta_{S_i} + \sum_{p=1}^{3} \lambda_{Mp} M_i^p + \sum_{p=1}^{3} \lambda_{Lp} L_i^p + \gamma_{\text{cohort}(i)} + \delta_{\text{gender}(i)} + \rho_{\text{age}(i)} + \kappa_{\text{comuna}(i)} + \varepsilon_i.$$

where Y_i is the outcome for student i, and S_i is the school attended by student i, as defined above. The coefficient β_{S_i} is a school fixed effect. M_i and L_i are the student's standardized fourth-grade math and language scores, respectively, which enter as third-order polynomials. The terms $\gamma_{\text{cohort}(i)}$, $\delta_{\text{gender}(i)}$, $\rho_{\text{age}(i)}$, and $\kappa_{\text{comuna}(i)}$ are fixed effects for the student's grade-8 cohort, gender, grade-8 age, and home commune, respectively. The error term is ε_i.

I also construct an unadjusted measure which corresponds to the raw school mean:

$$V_s^{\text{raw}} = \frac{1}{N_s} \sum_{i:S_i=s} Y_i.$$

We can partially see the role of these adjustments in Figure 2. Figure 2 shows the average value-added for math scores and STEM enrollment by region, comparing the unadjusted and adjusted metrics. For instance, schools in the capital region (RM) have very high unadjusted VA metrics, but adding controls significantly lowers these estimates on average. This indicates that these student populations tend to be positively selected.

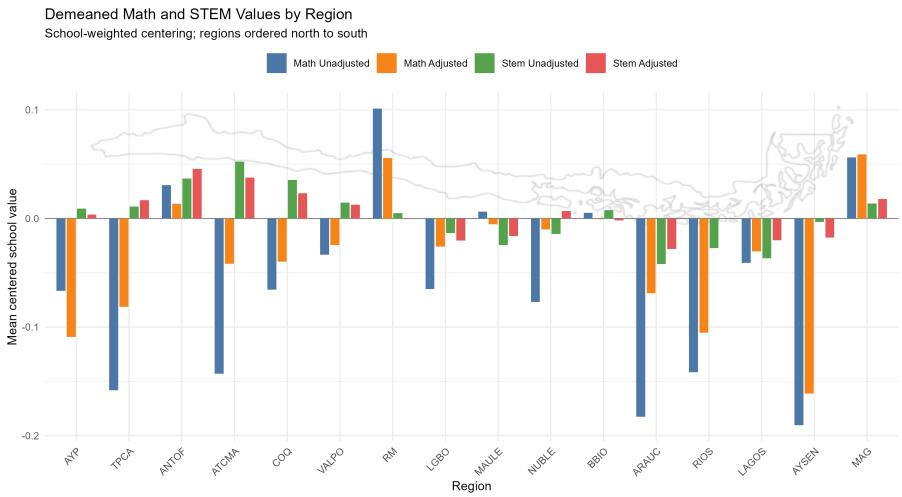

Figure 2: Average VA in math score and STEM enrollment, by region. Comparing unadjusted and adjusted metrics.

3.2 IV Regression: Projecting lottery-induced variation into Value-Added

Once we have the observational value-added for each school (β_s), I proceed as follows.

First, given the lottery design, I only have instruments for students who apply for a change of school in the centralized mechanism. More precisely, I reduce the IV estimation sample to students who participate in the grade-9 centralized assignment process on time and for whom the assignment-probability simulations imply non-trivial admission risk.

3.2.1 Students with assignment risk

I define a student as at (assignment) risk if the simulated assignment probabilities do not place all probability mass on a single option. Formally, student i is included only if

$$\max_{s \in \mathcal{S}} p_{is} < 1,$$

where p_{is} is the simulated probability that student i is assigned to school s.

3.2.2 School support

To reduce the dimensionality of the assignment-probability controls, I restrict the supported lottery school set to schools that appear with positive simulated assignment probability for at least 100 at-risk students. Let

$$N_s^+ = \sum_i \mathbf{1}\{p_{is} > 0\}$$

among students in the at-risk lottery sample. The supported school set is

$$\mathcal{S}^{100} = \left\{ s : N_s^+ \geq 100 \right\}.$$

While the number 100 is an arbitrary threshold, I present its relevance in Table 1. Table 1 reports how students in the broad grade-8 universe are distributed across schools by whether their attended school belongs to the SAE support set under alternative support thresholds. With this approach we can estimate the 2SLS for schools that span over 50 % of the population of students by using roughly 2000 controls (2 for each school, probability of assignment and indicator for zero probability).

Angrist et al. (2017) uses a threshold of 25 in their paper, but their number of schools is much smaller (lower than 50). Note that such a threshold would increase the number of controls by another 2000 to represent an extra 25% of the population. I intend to move in that direction but estimation is currently very computationally intensive.

Table 1: Student shares by SAE support threshold and school type

Threshold students at risk	School group	Schools	Students	Share
100	SAE <100	1,491	315,261	33.4%
100	SAE >=100	1,011	493,955	52.3%
100	Particular Pagado	497	87,373	9.3%
100	Other non-SAE schools	1,134	47,771	5.1%
25	SAE <25	530	89,266	9.5%
25	SAE >=25	1,972	719,950	76.2%
25	Particular Pagado	497	87,373	9.3%
25	Other non-SAE schools	1,134	47,771	5.1%

3.2.3 2SLS Regression

For each school-value measure β_s, I assign to each student the value of the school they attend:

$$D_i = \beta_{S_i},$$

where S_i is the school attended by student i. I instrument this attended-school value with the value of the school offered through the lottery:

$$Z_i = \beta_{O_i},$$

where O_i is the school offered to student i in the first round of the centralized assignment process.

The 2SLS system is:

$$D_i = \pi_0 + \pi_1 Z_i + P_i'\psi + X_i'\lambda + u_i,$$

$$Y_i = \alpha + \theta \widehat{D}_i + P_i'\psi + X_i'\lambda + \varepsilon_i.$$

Here Y_i is the student outcome of interest, D_i is the observational value of the school attended by the student, and Z_i is the observational value of the school offered by the lottery. The vector P_i contains assignment-probability controls, and X_i contains additional student-level controls. The parameter of interest θ measures the causal effect of a lottery-induced increase in the observational school-value index.

4 Preliminary Results

4.1 Core Results

Table 2: Scalar school-value IV estimates

	Math		Language		STEM	
	Unadjusted	Adjusted	Unadjusted	Adjusted	Unadjusted	Adjusted
θ	0.208	0.431	0.167	0.393	0.345	0.584
SE	0.033	0.057	0.032	0.062	0.046	0.068
p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001

Table 2 shows the results of the 2SLS regression above for each of the three outcome variables, comparing both the adjusted and unadjusted value-added metrics.

Overall, this allows us to conclude that the VA metrics are predictive of lottery-induced causal effects across specifications. The fact that the adjusted estimates are larger suggests that adjusted value-added metrics are better at predicting the causal effects of schools on these outcomes, in line with prior literature.

References

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